Rinse surfaces: environment report

Environment `surface-rinsing` (package `surface_rinsing/`). Robot: Franka Mobile FR3 Duo (default) or its single-arm variant, on the stack qualified in the `classroom_arrangement` environment with the contact refinements of `painting`. Rigid physics: Newton 1.5 with MuJoCo on the CPU at 0.5 ms. Water: a Warp thin-film and packet engine on the GPU, written for this environment. Rendering: an Isaac Sim 5.1 RTX path-tracing worker that never steps physics.

This report says what was built, on what basis, what was measured, and what is not right yet. Every number comes from a run of the environment's code. "Assisted" always means a labelled phantom force, torque or state write that no policy action can reach.

Contents: 1 the activity and what was built · 2 the room · 3 the robot · 4 water as physical state · 5 mechanisms (valves, hoses, sponge, grip) · 6 numerics and cost · 7 parameter ledger · 8 demonstrations

1. The activity and what was built

The action group is rinsing: water, by itself or after a wash, carrying soap, loosened soil and debris off a surface, with the worker choosing where the water goes, how long, and what happens to it afterwards. Its weight sits in commercial kitchens and janitorial work. The built scenario is one room where four versions of that work happen every shift:

- **the three-compartment sink** (wash, rinse, sanitize, air-dry: Food Code 4-603.16): glasses and dishes come out of the wash water coated with a 15-40 um film of detergent solution and patches of suds, must be rinsed on every face, held in the sanitizer for 30 s and left to drain; a pan with dried egg is scrubbed with a sponge first; the sponge itself is rinsed and wrung;
- **the pre-rinse station**: plates with dried tomato paste are blasted with a 1.15 gpm spray valve before the dish machine (the ASTM F2324 cleanability test is exactly this);
- **the prep sink**: produce is rinsed under running water in a colander;
- **the can-wash bay**: a scrubbed garbage can and the floor are hosed down toward a floor drain.

What makes these tasks is physical, and each item below exists in the simulation because a task turns on it:

What a worker deals with	What the simulation does about it	Where shown
water comes only when a handle or lever is worked, mixes hot and cold, pushes back on the hand	supply network solved every tick from the fixtures' joint angles; jets are packets with momentum; the reaction acts on the valve body	§5.1, B3, D1
a rinse is a coverage problem: any face the water reaches is clean in seconds, a face it misses stays soapy	surfactant is a per-cell quantity (in solution, adsorbed, in foam) carried by the film flow; gauges read every face	§4, A1, P10
the inside of a glass is reached through its mouth, or by flooding	cavities are pools with level tables; a dunked glass floods by its mouth and exchanges with the bowl	§4.4, A1
a spray jet cleans where it hits and splatters everywhere else	impingement soil law under the jets; Bhunia-Lienhard splatter leaving as drops that wet the robot, the shelf, the floor	§4.2, B3, B6, A6
dried soil comes off only after it has soaked	hydration state per cell, strength falling with it	§4.5, B6, A5
a sponge holds soapy water and gives it up when squeezed	two-body sponge with a foam spring and a pore pool	§5.3, P15, A5
a soapy glass slips in the fingers	finger-pad friction read from the film under the pad	§5.4, P17
a can full of rinse water is heavy and must be tipped	the pooled water is mass on the body and pours over the lip	§4.4, A4
water runs to the drain, or onto the floor	sloped charts, drain cells, run-off from everything else, named accounts; a ledger that closes	§4.6

Not built: the potato-scrubbing task and its vegetable brush (V2, E12), the hotel-pan variant (R1b), flatware in the dishware task, grease that needs detergent, slosh of carried liquid as its own mode, and bedload transport of grit (E10).

2. The room and its things (`layout.py`, `ware.py`, `scene.py`, `render.py`)

World frame: x east along the north wall, y north, z up, floor at z = 0. The room is 7.2 x 5.4 m with a 3.05 m ceiling. Dimensions are in `params.py` with their sources (§7).

- **Pot sink** (north wall): three 18 x 24 x 14 in bowls, rim 0.94 m, two 24 in drainboards pitched 6 mm toward the bowls, 10 in backsplash (Advance Tabco FS-3-1824-24RL spec sheet). Wash and rinse bowls hold 70 L at 43 C, the sanitizer 70 L at 24 C; the wash water carries 0.75 g/L of surfactant and 2 cm of suds. Each bowl has a lever waste on the apron. A wall-mounted pre-rinse unit stands over the wash/rinse partition: hot and cold wrist blades, a 12 in swing spout with its own quarter-turn valve (2.2 gpm laminar outlet), and a B-0107 class spray valve hanging from an overhead spring on 0.63 m of free hose. A wall shelf over the clean drainboard takes the inverted pan.
- **Pre-rinse station** (west wall): a soiled dish table with a 20 x 20 in sink under a peg rack on rails, the same kind of pre-rinse unit, the dish machine's hood and a clean table beyond it.
- **Prep sink** (east wall): one bowl with a drainboard, a wall faucet with a turned-down spout, a work table with a steel bowl and a cutting board.
- **Can-wash bay** (south-east corner): quarry tile sloped 1 % to a 6 in floor drain, an open hose reel 1.5 m up with 2.5 m of hose paid out and a squeeze valve docked head-down in an open-bottomed wall holster (a sleeve whose ring floor seats the valve's bumper and lets a docked valve's spray fall through to the floor), a 32 gal can (0.69 m tall, 4.5 kg), a wire dunnage shelf (clean storage that must stay dry).

Free bodies: rocks glasses (0.32 kg) and mixing glasses (0.42 kg), 9 in plates (0.47 kg), soup bowls, a 10 in aluminium fry pan with a riveted handle (0.84 kg), a 5 qt perforated colander, fifteen cherry tomatoes (11 g), a steel bowl, a cutting board, a 25-compartment glass rack and a peg rack, the can, the scrub sponge. Revolved ware is built from one profile per item: the same polygon gives the convex collision staves (so a finger fits inside a glass), the film chart with its regions (inside, rim, outside, foot / face, back), the cavity's level table and the drawn mesh. Fixtures with joints (blades, spout, spray valves on their hoses and balancer, the reel hose, lever wastes) are Newton articulations; everything else of the room is static geometry.

Appearance: three Poly Haven textures (floor tile, wall tile, block wall) and eleven dressing models are pinned by hash in `visual_assets.json` (fetched once to an untracked cache); steel, chrome, glaze, glass and water are MDL materials; what is wet, soapy or soiled is drawn from the cells' state on every frame (soil and suds as opacity-mapped overlays, pools as dielectric volumes, water in flight as curves).



Figure 1. The room at the start of four tasks, from fixed room cameras. Top left: the pot sink in rinse-glassware, with the washed glasses on the soiled drainboard and the glass rack on the clean one. Top right: the pre-rinse station in pre-rinse-plates, with the soiled plates in the peg rack. Bottom left: the prep sink in rinse-tomatoes, with the colander of tomatoes, the steel bowl and the cutting board. Bottom right: the can-wash bay in rinse-garbage-can, with the can, the hose reel, the hose valve in its holster and the dunnage shelf.

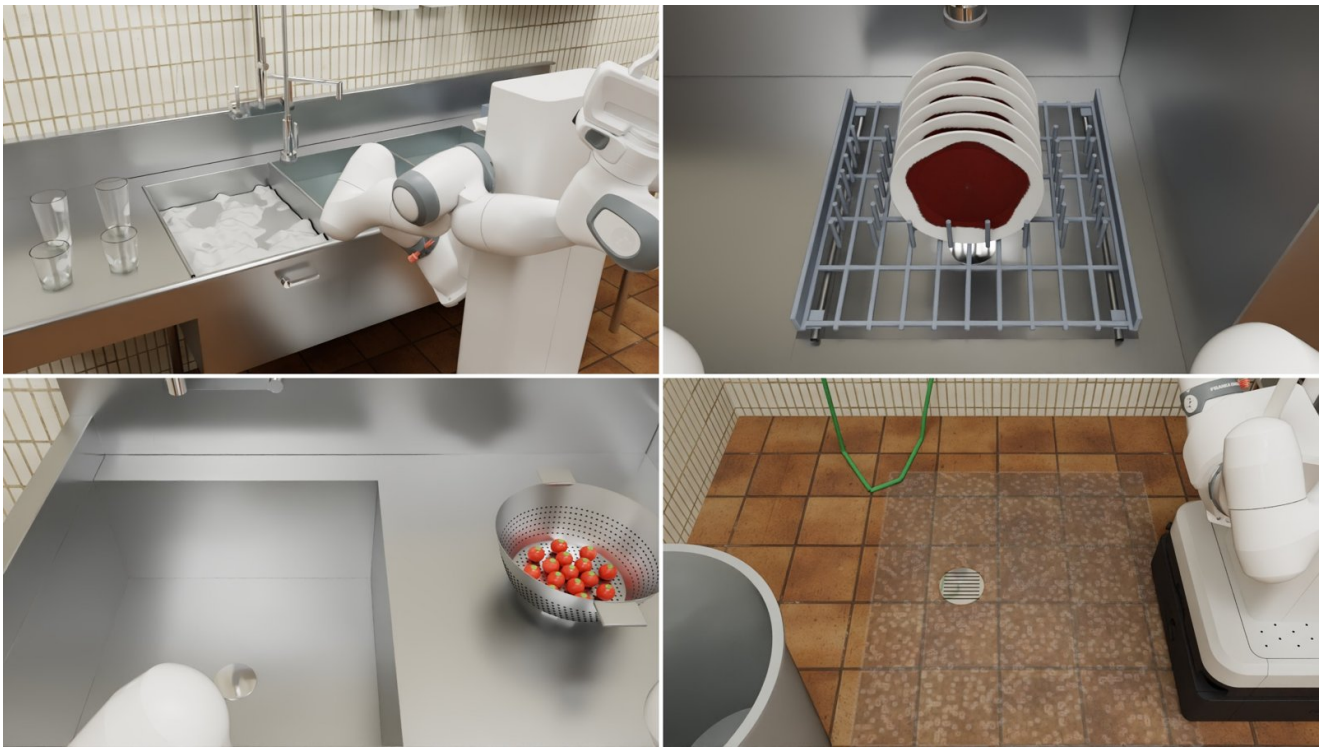


Figure 2. Start states from closer fixed cameras. Top left: rinse-glassware, the glasses beside the wash bowl and its suds. Top right: pre-rinse-plates, six plates with dried tomato paste in the peg rack. Bottom left: rinse-tomatoes, fifteen cherry tomatoes in the colander. Bottom right: flush-floor, the debris on the wet bay floor round the drain. Soil, suds and water are drawn from the cells' state.

3. The robot (`robot_setup.py`, `robots/catalog.py`, `robots/rigs/rinsing-mobile-fr3-*.json`)

The environment takes its robot from the shared catalog. `mobile-fr3-duo` (default) and `mobile-fr3-single` are the rigs `rinsing-mobile-fr3-duo` / `-single`: one instance on the `floor_start` mount with the `classroom` controller profile, the stack qualified in the `classroom_arrangement` environment (swerve base with wheel slip, Spine lift, FR3 joint servos with gravity compensation and speed limits, Franka Hand in width or force mode at up to 70 N per finger, collision reflexes that latch a protective stop, head, wrist and base cameras, IMU and LIDAR). Nothing in that stack was changed; `world.py` adds the water coupling, soap-dependent pad friction, the no-slip hold setting of §6.3 and rolling resistance for the tomatoes.

The robot starts at its station with the arms stowed, the shoulders 7-11 cm short of the fixture's front edge, and the Spine raised (0.45 m at the pot sink and the pre-rinse table, 0.42 m at the prep sink, 0.25 m in the can bay) so that the head camera, which rides on the Spine's carriage, looks down into the work (Figure 3). Every start is checked for overlaps: over seeds 0-4 of every task no contact is deep at tick 0 and the settle latches no protective stop (P13). Tools start in their real parking places (spray valves hanging, the hose valve in its holster, the sponge on the ledge), not in the hand.



Figure 3. The head camera at the start of four tasks, with the Spine raised: the pot sink's three bowls, the plates in the peg rack, the colander on the prep sink's drainboard and the garbage can in the bay. The white shapes at the bottom edge are the robot's own arms.

Refused before anything is built, each with its reason:

- `mobile-fr3-single` on `rinse-garbage-can`: tipping a can that holds litres of rinse water takes two hands or more than one FR3's 3 kg payload;
- `g1`: the water system is coupled to the Mobile FR3 world, and no standing-at-a-sink posture is qualified for the G1 in this repository. The decisive geometry was measured rather than asserted (forward kinematics of the pinned URDF): standing upright at the front edge of each fixture, a G1 palm reaches the bowl floors, the hanging spray valve, the plate rack and the can's rim, but not the faucet blades of any station (4-15 cm short), the sponge's ledge, the wall shelf or the can's floor; bowing 40 deg from the hips brings the pot sink's cold blade, the prep sink's blade and the sponge's ledge into reach kinematically and still leaves the pot sink's faucet valve 9 cm, the pre-rinse hot blade 3 cm and the wall shelf 0.67 m away. A G1 port therefore needs a balanced bowing or stepping controller, which does not exist here;
- `dual-fr3`, `single-fr3`, `dual-panda`: no pedestal layout is built, and one pedestal cannot serve four stations.

4. Water as physical state (`water/engine.py`, `surfaces.py`, `pools.py`, `system.py`)

4.1 Representation

Water is in exactly one of five stores, and the ledger of §4.6 sums them every time it is read:

1. **in flight** as packets (a ring of 262,144 on the GPU): position, velocity, mass, diameter, kind (coherent jet segment or drop), six species;
2. **films** on charted surfaces: every surface that matters to a task is a chart of cells (2-10 mm) attached to its body: rectangles (drainboards, tables, floor), revolved profiles (ware, the can), ellipsoids (tomatoes). Each cell carries depth h, discharge q = h u in its own tangent plane, and the species per area;
3. **pools**: the sink bowls (boxes), the cavity of every container (a level table from volume samples, re-read as the body tilts), the sponge's pores;
4. **edge accumulators**: water that has left a film at a lip or an open edge and not yet formed a drop;
5. **sinks**: named accounts for water that reached something with no chart (robot links, fixtures and walls, racks, the dunnage shelf, the floor outside the bay, the building drain).

Species: surfactant in solution, suspended soil, a contamination tracer, heat (h T), foam liquid and the surfactant in it. Adsorbed surfactant, adhered soil with its hydration, and the soil's tracer flag stay on the cell.

4.2 Jets, impact and splatter

An outlet emits one packet per orifice per 0.5 ms tick with mass rho Q dt / n and velocity v = Q / A_eff along the orifice axis plus the outlet's own velocity (§5.1). Jet packets fly ballistically; a coherent jet contracts by continuity and breaks into drops of 1.89 d beyond the capillary length L_b = 12 d sqrt(We) (1 + 3 Oh) (Weber, as quoted by Bhunia & Lienhard); drops feel Schiller-Naumann drag. Each tick a packet's swept segment is tested against every body's meshes (a BVH per body) and the pools' free surfaces; the first front-facing hit wins.

On a charted cell, with w the packet's velocity relative to the surface and n its normal:

- **force**: the normal impulse m (w.n) n / dt and its moment go to the struck body. Summed over a jet this is the momentum flux rho Q (w.n): the "spray force" of the DOE standard;
- **splatter**: a fraction xi of the packet leaves as two drops. For jets xi(We_d, l/d) is the Bhunia & Lienhard (1994) correlation with its onset l/d = 130 / (1 + 5e-7 We^2) and the plateaus read from their Fig. 8 (0.30, 0.37, 0.75); falling drops splash only above We Oh^-0.4 = 2100 (Cossali et al.). The drops take an equal volume of the impact cell's film with them (so splatter carries soap and soil), leave from a ring of 3 jet diameters at 15-45 deg above the surface with 0.27-0.63 of the jet's normal speed in a random azimuth, and keep the jet's tangential velocity. The last point is a Galilean argument (in the frame sliding along the wall the impact is normal) and makes an oblique jet throw its splatter forward; the speed and angle ranges are estimates (§7);
- **film source**: the rest enters the impact cell and its two rings of neighbours with the jet's tangential velocity plus a radial outflow at |w.n|;
- **impingement pressure**: 1/2 rho (w.n)^2 over the jet's share of the cell area feeds the soil law of §4.5.

A hit on a pool's surface or wall adds to the pool; a hit on anything without a chart runs off it or stays in its account (§4.6).

4.3 Films

On each chart the engine integrates the depth-averaged shallow-film equations in the body's frame

```
dh/dt + div(h u) = S_h
d(h u)/dt + div(h u u) = -g_n h grad h + h g_t - tau_w / rho + S_q
```

with g_eff = g - a_P the effective gravity at the cell (the body's acceleration low-passed over 8 ms, plus Coriolis from its rotation), g_n and g_t its normal and tangential parts, and wall friction tau_w = 3 mu u / h (laminar) or the Blasius form above Re_f = 1000, integrated implicitly so thin films relax to lubrication drainage u = g_t h^2 / 3 nu exactly. Viscosity follows temperature (Vogel fit: 1.00 / 0.62 / 0.46 mPa s at 20 / 43 / 60 C).

The discretisation is finite-volume on the cell graph: donor-cell upwind fluxes of mass, momentum (rotated from the donor's tangent plane into the receiver's) and species; a Rhie-Chow face velocity

```
u_f = 1/2 (u_a.n_a + u_b.n_b) - D_f g_n [ (h_b - h_a)/delta - 1/2 (G_a.n_a + G_b.n_b) ],   D_f = 1 / (1/dt + 3 nu / h^2)
```

with G the Green-Gauss depth gradient, which removes the one-cell checkerboard a collocated scheme otherwise supports; and a per-donor limiter that keeps depths and species non-negative. Depth, species and every transfer are float64.

Closures on top of the flow, each with its basis in §7:

- **lips**: at a sharp convex edge the film separates as a sheet when its speed exceeds the Duez et al. (2010) inertial-capillary threshold U_c^2 = K (1 + cos theta) (sigma / rho) (r_i + e/2) r_i / e^3 and otherwise wraps round to the far face (the teapot effect: how soapy water reaches a plate's back);

- **drips**: a film facing down sheds drops above a drip depth; water crossing an open edge leaves as drops of >= 4 mg;
- **beads**: on a non-wetting substrate a film thinner than what pinned drops can hold stays put. The depth is h_wall / sqrt(sin alpha) for slope alpha (hysteresis against rho g h sin alpha), at most 5 h_wall near level, with h_wall = 0.10 mm on polymer and tomato skin, 0.06 mm on steel and aluminium, 0.04 mm on tile and 0 on glass and glaze (they sheet);
- **adsorption**: Langmuir exchange between the film and the surface, d Gamma/dt = k_a c (Gamma_m - Gamma) - k_d Gamma;
- **foam**: suds are their own liquid store on a cell with a yield stress (Mason et al.: 5 Pa), a 300 s half-life, and a fast collapse where the liquid under them is below half the CMC;
- **withdrawal**: a surface leaving a pool carries a Landau-Levich film of the pool's liquid, and up to 3 mm of the pool's surface foam.

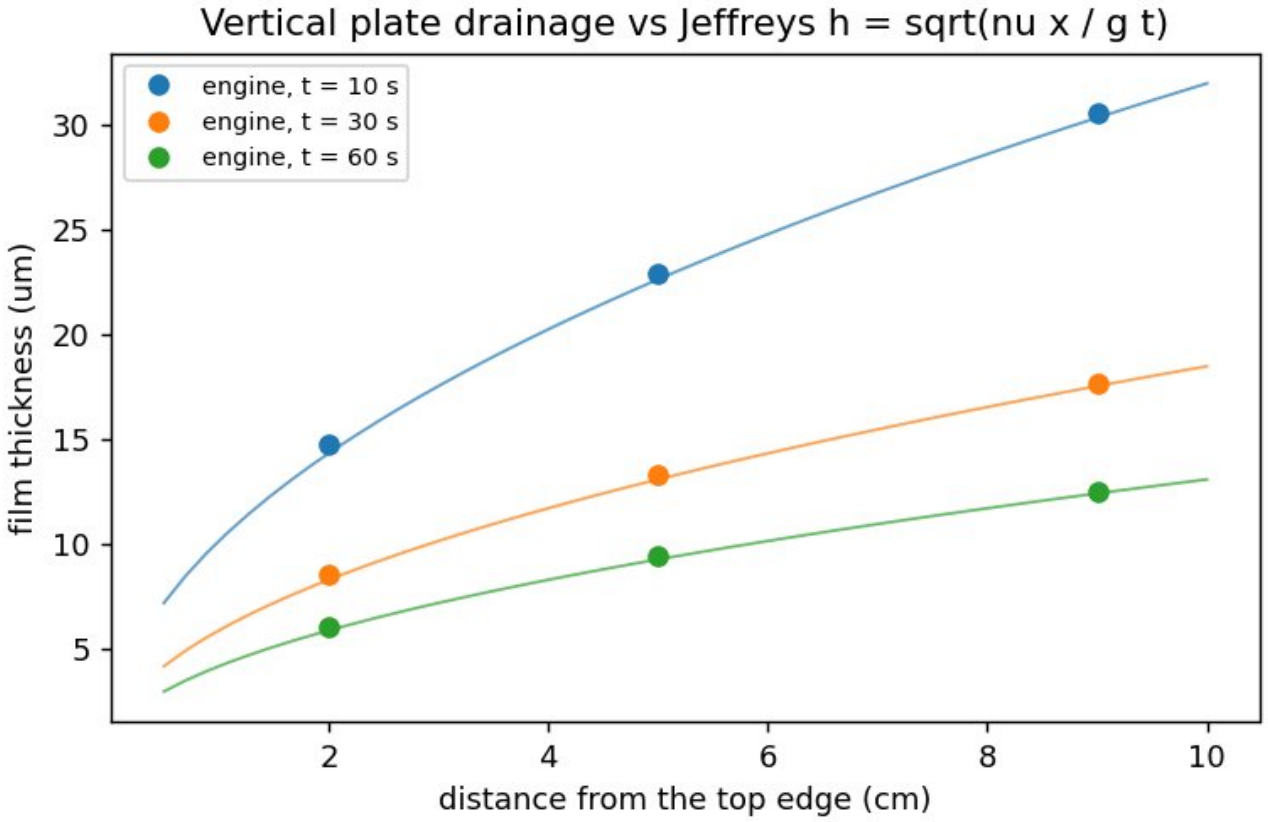


Figure 4. Bench B2: a film draining down a vertical glass plate, 2 mm cells. Engine (points) against Jeffreys' lubrication solution h = sqrt(nu x / g t) (lines) at 10, 30 and 60 s. The engine's film is 0.8-3.2 % thicker than the solution at every point. Against a run at 1 mm cells and half the step, the drained thickness differs by at most 1.4 % at 2 mm cells and 3.9 % at 4 mm cells.

4.4 Pools

A static bowl's level is (V + V_displaced) / A; its lever waste drains Q = C_d a f sqrt(2 g h) with f the lever's opening; above the rim it overflows over the front. A cavity's free surface is the plane normal to the effective gravity that holds its volume, from a table of volume samples re-sorted when the body tilts; it spills over the lowest point of its rim as a weir, and the colander loses Q = C_d a sqrt(2 g depth) through every hole below the surface. Whatever leaves a pool re-enters the packet ring at the rim or hole with the body's velocity there. Pool foam decays by the same law as film foam and leaves first with any overflow.

A container dipped in a bowl floods: the water inside rises toward min(outside level, highest rim point) at the weir rate of the submerged mouth, so an upright or sideways glass fills while a mouth-down glass keeps its air; a flooded cavity exchanges with the bowl at 0.05 + 2 |v| per second (estimate), which is how dunking and swishing dilute soap into the rinse water. Cells below a pool's surface give their film to the pool, come to adsorption equilibrium with it, soak at the pool's temperature, and erode by the body's motion through the water (skin friction with c_f = 0.005). Buoyancy and drag on free bodies come from volume samples, including trapped air. Pooled and film water is added to its body as mass at its own centre, so a can holding 2 L is 2 kg heavier and a full glass tips differently.

Film standing on the colander's perforated sheet above the 60 um that clings runs into the holes (0.1 s time constant) and leaves with the cavity's water.

4.5 Soil

Three soils exist: dried tomato paste on the pre-rinse plates (0.20 kg/m2 over the 7 in flat: the F2324-03 recipe), dried egg on the pan (0.08 kg/m2), and dust or debris on the tomatoes, in the can and on the bay floor (up to 2 g/m2). Each soiled cell carries mass, a hydration w and a strength

```
sigma_c(w) = sigma_dry (1 - w)^2 + sigma_wet,      dw/dt = (1 - w) / t_soak(T) while wetted,  t_soak halves per 15 C
```

and loses mass by four routes, the first three only while wet:

- impingement: dm/dt = k_i (p_imp / sigma_c - 1)+ over the jet's share of the cell;
- film shear: dm/dt = k_e (tau_w - sigma_c)+;
- a cleaning front in the manner of Wilson et al. (2014), advancing from clean cells at k' M with M the film's momentum flux per width;
- abrasion under the sponge: dm/dt = k_abr p v_slip sigma_ref / sigma_c (Archard form), §5.3.

Removed soil becomes suspended solids in the film (and tracer, for soils flagged as contamination) and travels with it. The paste constants are from bench B6 (Figure 5); the egg and dust constants are estimates.

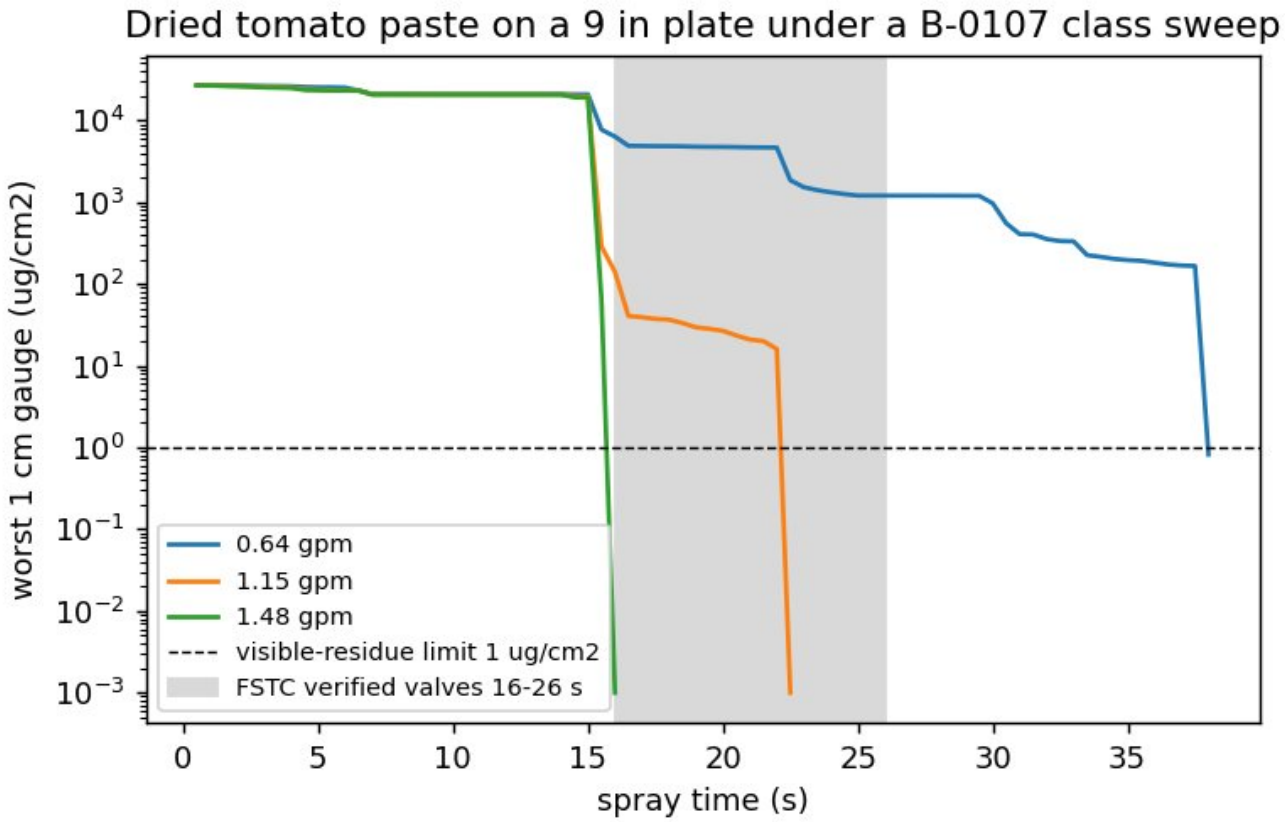


Figure 5. Bench B6, the ASTM F2324 cleanability test: the worst 1 cm gauge on a 9 in plate with 5.07 g of dried tomato paste, under a B-0107 class valve at 60 psi swept in a raster 0.20 m from the plate (0.30 m/s, rows 2 cm apart), at three flows. The soak time t_soak is calibrated at 1.15 gpm against the FSTC range for verified valves, 16-26 s (grey): the plate is clean of visible residue (every gauge under 1 ug/cm2) after 22.5 s. With the constants frozen, 1.48 gpm takes 16 s, inside the range, and 0.64 gpm takes 38 s, outside it. The held-out prediction fails at the low flow.

4.6 Where the water goes, and the ledger

Films reaching a drain cell (the floor drain's strainer, a bowl's waste) go to the building drain. Films on the room's charted surfaces (the floors, drainboards and tables) do not run under an item standing on them: every 10 ticks the cells under each upright item's standing ring (its profile within 2.5 mm of the standing plane, within 4 mm of the surface) are marked, the face flux into and out of them is zero and a jet's impact spreads none of its water into them, so water runs round a can, a glass or a plate. Without this the bay floor's film, and the debris in it, would run under the parked can, where no jet can reach. Impact water spread into the footprint's edge could never leave it, and drops running off the can's wall into the edge cells, with the dry footprint counted in the depth gradient of the open cells beside it, would pile debris into single cells. Faces to covered cells are therefore walls for the gradient as well, and a drop landing on a covered cell goes wholly to the open cells round it.

Colliders that carry no film chart (the robot, hoses, valves, levers, rack wires, table frames, walls, the shelf) have named accounts. What strikes one transfers its normal momentum to the body and then, by where it struck:

- a thin moving part (hose, valve body, lever, blade, swing spout, rack wire) sheds all of it: the water continues as a drop from the point it struck and passes that part;

- the robot and the static fixtures shed what strikes a face steeper than 60 deg from the horizontal or overhanging: the drop starts 3 mm off the face at the face's velocity and falls;
- faces that look up (a ledge, the top of the base), the dunnage shelf and the floor outside the charts keep it in their account. A holder that should pass water must therefore have its hole: the holster's floor is a ring, and the spray of a valve squeezed while docked reaches the bay floor. In the stress run on rinse-garbage-can (P11 chaos), with the hose valve squeezed at 30 N for 20 s, the room's static parts, the holster among them, kept none of the 4.19 kg sprayed. The shelf's and the floor's accounts are what the `contained` gates read.

Thin moving parts keep nothing: in the floor flush (A6) 1.08 kg of water struck the hoses, valves and levers (the `tools` account, which holds the reel's hose lying under the nozzle) and they kept none. `totals()["struck"]` reports what struck each account, `["sinks"]` what it kept. Water kept on a ledge does not run on.

The ledger compares, in float64, water + emitted, surfactant + emitted and solids against their values at reset; `no_simulation_fault` fails any state whose water, detergent or soil residual exceeds 1e-9 of its total (or 1 ng when there is almost none). Water, read once per transfer, stays exact. The species need three more rules:

- where a splattering packet takes film with it, the cell's detergent and soil are read once, for the drops and for the subtraction. Read a second time, after other packets in the same GPU launch have changed them, they are created in single ticks;
- splatter draws on a per-cell budget (the film at the start of the tick, taken atomically), so the takes can never exceed what is there, and the cell update counts what its clamps create. Without it, when the hose first hits the soapy can, several packets strike one cell in a tick and each takes up to half of the film it read. Together they take more detergent than the cell holds, and the clamp to zero at the cell update creates the difference;
- pools use the same kind of budget: the submerged walls' adsorption and the emerging films draw atomically on the volume, detergent and foam the pool held at the start of the tick. Each cell emerging from a pool carries out a Landau-Levich film. If each only checks that the pool holds 20 times its own film, many cells emerging in one tick can take a small pool, such as a splash in the parked can, below zero in volume and detergent. The ledger still closes, because the negative pool balances the films. When clean water then lands in it, the pool's concentration goes negative, the films it coats carry negative detergent, and the clamps create detergent.

With both budgets in place the flush-floor demonstration's detergent ledger closes to 6.2e-12 (A6). Every verdict reports the clamp count (across the eleven judged runs of the assisted and robot demonstrations the clamps created at most 1.1e-17 kg of water, 3.3e-26 kg of detergent and 4e-20 kg of soil). Across the same runs the ledgers close to 1.7e-10 (water), 1.2e-10 (detergent) and 2.2e-10 (soil) at worst, all three in the can rinse (A4). The evaluator's negative controls (P10) include two for the gate (10 ug of detergent or of soil from nothing).

Where a drop is born. Packets collide with the front faces of the colliders only, so a drop created inside a solid passes through it. Every place a drop is created therefore casts its birth offset against the colliders. The ray starts 0.3 mm behind the birth point (under the thinnest wall, the colander's 0.8 mm), so a point lying on a face still sees it, and the drop stops 0.5 mm short of the first face crossed. No drop is born below 0.5 mm above the floor plane, which is the room's lowest surface everywhere. The births covered are a splatter crown's base and rim, a drop shed off a steep face (including the hose, which lies pressed into the floor), a drop leaving a film's edge (born on the film's side of the edge) and a pendant drip. Water re-entering from a pool (over a rim, through the colander's holes) or from the sponge is tested from 1 cm above its birth point against the room's installed parts and placed above any of their surfaces in between. Moving bodies are left out of that test: such water is born outside its own container by design. If they were tested too, the outflow would be put back inside, and the colander could not drain nor the tipped can empty.

Without these rules the crown's ring (3 jet diameters out from the hit) puts drops born within a centimetre of a wall inside it, and the foot of the can's wall releases edge drops and drips below the floor, where they fall out of the room. The water ledger still closes, because water leaving the room is counted as lost. While the can is leaned back 45 deg on its lip for the underside spray, its pool overflows at the lip, under the floor. Hence the guard on pool outflows and the floor plane rule. Drops falling past the bay's open edge are sometimes not met by the catch plane under the floor (two large triangles at -1.5 mm), and the cause was not found. A segment crossing that plane unstruck is therefore kept there analytically. As a safety net, any packet found under the catch plane is lifted above the floor. Every verdict counts what each of these did: across the eleven judged runs the lift net under the floor never fired, the floor plane kept at most 0.71 g that crossed it unstruck, and 11.6 mg of water left the room, both in the can rinse (A4). Lost water is counted by cause (aged out, left the room, overwritten in the packet ring) and reported in every verdict (`no_simulation_fault.detail`).

5. Mechanisms

5.1 Supply and valves (`water/network.py`)

Every outlet solves its branch from the static supply pressure each tick:

```
p_s = f_D (L/D) rho v^2 / 2  +  rho/2 (Q / A)^2,     1/A^2 = 1/(C_d A_mix (f_h + f_c))^2 + 1/(A_o g)^2
```

with the Darcy factor 64/Re or 0.316 Re^-1/4 for the line, the mixing cartridges' opening fractions f_h, f_c read from the wrist blades' joint angles (5 deg dead band, linear to the quarter turn), the outlet's own gate g (the squeeze lever through a smoothstep between crack and full travel, or the spout's valve), and the orifice face's effective area A_o = Q_rated / (C_v sqrt(2 p_rated / rho)). The swing faucet is capped at 2.2 gpm by its laminar regulator. The mixed temperature is the flow-weighted mean. The result reaches the outlet through a 2 ms lag.

The B-0107 class valve (1.15 gpm at 60 psi through nine 0.60 mm jets at 28 m/s) and the reel's EB-0107 class valve (3.89 gpm rated; 3.3 gpm at 24 m/s and 5.1 N through the hose's Blasius loss, from cold water behind a wall valve left open; the demonstrations measure 3.33 gpm at 24.2 m/s and up to 5.04 N, A4 and A6) have a lever on a revolute joint with a preloaded return spring applied as an explicit torque: 13 N at the pad cracks it, 22 N opens it fully over 10 mm (US 7,150,415). The hinge carries 1e-3 kg m2 of added rotor inertia (0.28 kg at the pad, so that a 20 g lever is well-conditioned against finger contacts) and is critically damped against the spring (0.12 N m s/rad; response 17 ms). Held part-way by the robot's 40 N grip (D1), the lever sits at 0.114 rad and the valve passes a steady 0.83 gpm. A hinge of 0.05 kg m2 and 0.02 N m s/rad is a 14 kg lever ringing at 1.35 Hz, and a part squeeze then hunts between no flow and nearly full flow. The momentum each tick's packets carry is applied to the valve body as their reaction. Nothing opens a valve except its joint angle; a closed valve emits exactly zero. The spray valve delivers 1.147 gpm at 28.1 m/s with up to 2.02 N of reaction on the hand (A2, D1), and its jets strike a plate 0.2 m away with 2.035 N, the momentum flux to 0.02 %, against 2.1 N in DOE's spray-force class for this valve (B3).

5.2 Hoses, the overhead spring and the reel (`scene.py`)

The pre-rinse valve hangs on nine 7 cm capsule links with two bending axes per joint (EI 0.05 N m2) from a slide with 0.6 m of travel carrying an explicit spring F = m g + 30 N/m s and 8 N s/m of damping: the unit's overhead spring and 44 in hose reduced to a hose on a balancer. The reel is a static drum with the last 2.5 m of 3/8 in hose as a chain of 12.5 cm links lying on the floor and rising to the valve in its holster; the reel does not pay out or rewind.

5.3 The sponge (`water/sponge.py`)

Two bodies, the scour pad with the lower cellulose and the upper cellulose, joined by a slide of 10.5 mm. The foam spring F = E A_eff (eps + 3 eps^3) with E = 50 kPa over 30 cm2 is an explicit force; the slide's damping (30 N s/m, water forced through the pores) is the joint's own, integrated implicitly. The pores are a pool of capacity phi A H_c (1 - eps) = 79 mL uncompressed: water striking the sponge enters it, whatever exceeds the capacity leaves (as drops from the lower face, or onto the surface it presses on), and immersed it fills and exchanges with the bowl. Pressed on a charted surface, the sponge spreads the summed normal force of MuJoCo's contacts as a uniform pressure over every cell within 6 mm of its 111 x 66 mm block and facing it, abrades soil there by the Archard law (cells nearest the pad four times those nearest the cellulose) and takes up film by capillarity while under capacity. The 6 mm is the cellulose's softness: the colliders are the block's firm core, 6 mm inside it all round (E = 50 kPa over an 8 cm2 end and 111 mm of length is 0.36 N/mm, so an end pushed into a corner gives by centimetres under a few newtons), and the drawn sponge and the contact patch are the full block. The core's ends are arcs of 40 mm radius in plan, not square: a real sponge's corners are its softest part. A square core against the pan's 97 mm wall has its corners 3-4 mm inside the wall's 4 mm convex sector hulls. The solver pushes them out sideways through the sectors' side planes at no applied force, and sponge and pan lock together. The pore water's mass (up to 79 g on an 8 g sponge) is lumped into the base with its own inertia, as a uniform block filling the 12 mm of cellulose, because capillarity holds it and it turns with the sponge. The lumping used in `painting` adds only mass and leaves out the liquid's inertia, which suits a pool that does not turn with its container. With it the wet sponge turns like the dry one, with about a tenth of its real inertia, and under a scrub's friction it rocks until an edge goes through the pan's floor hull. With the inertia in, the scrub demonstration (A5 scrub: 57 s of strokes at 15 N over the middle and 25 N round the edge) runs with no fault and no protective stop. With rigid full-size faces the sponge cannot reach the foot of a pan's wall at all: a straight edge bridges a round corner as a chord and a flat face cannot enter a concave one.

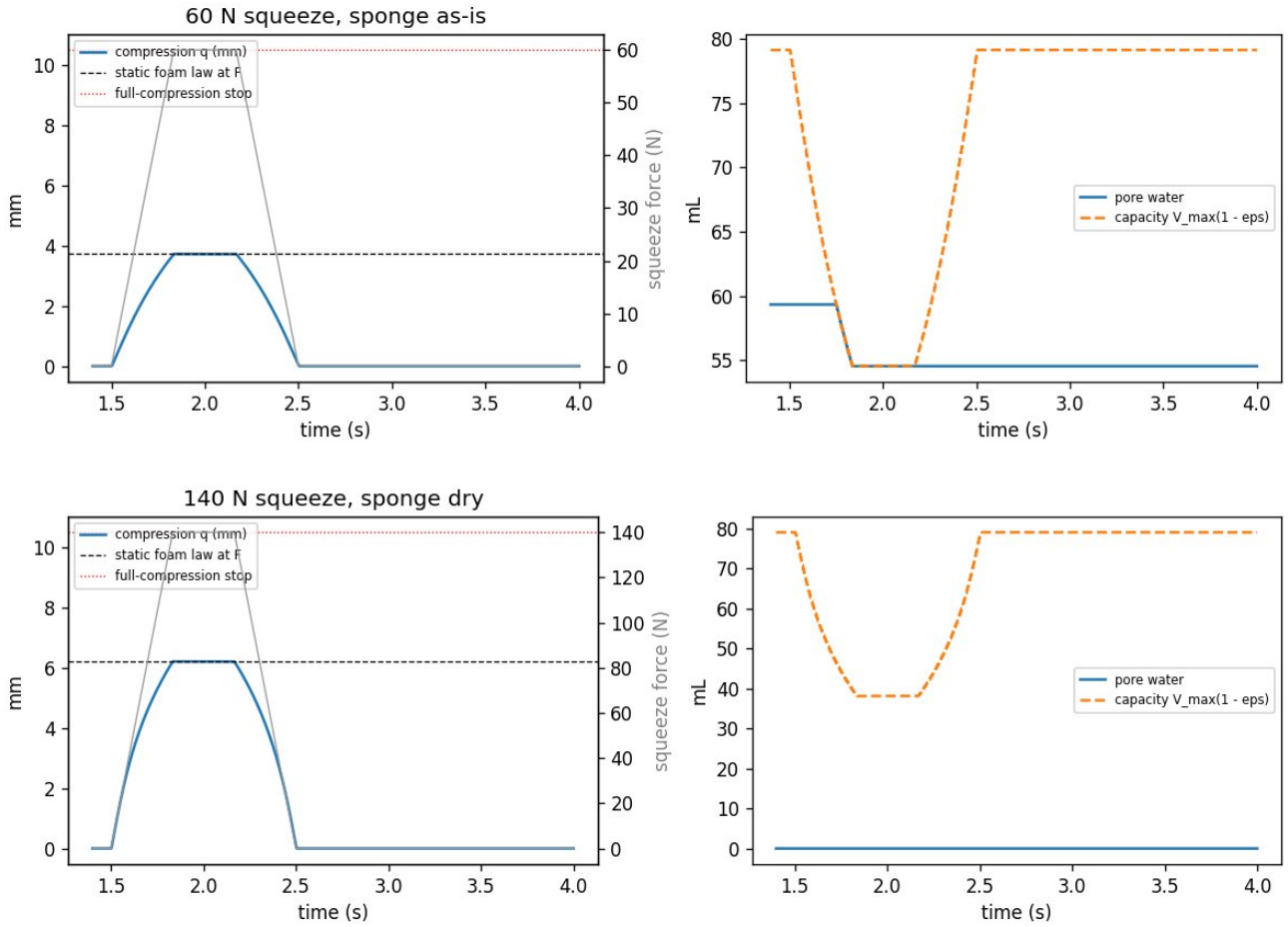


Figure 6. Probe P15: the sponge held by a phantom hand and squeezed across its thickness by a pure force pair, ramped over 0.33 s, held 0.33 s and released over 0.33 s. Top: the wet sponge (59.3 mL in its pores) at 60 N. Bottom: the dry sponge at 140 N, twice the hand's 70 N per finger. Left: the slide's compression against the static foam law at the applied force (peak 3.727 mm against 3.724 mm at 60 N, 6.214 mm against 6.210 mm at 140 N) and the force (grey). Right: the pore water and the squeezed capacity V_max (1 - eps). At 60 N the capacity falls to 54.5 mL. The wet sponge expels the water above it and holds 54.55 mL after release.

5.4 Grip on wet and soapy ware (`water/system.py: _pad_friction`)

Every 10 ticks each finger pad's friction coefficient is set from the cell under its contact: mu = mu_wet (1 - phi_s) + mu_soap phi_s on a wet cell and mu_dry (1 - phi_s) + mu_soap phi_s on a dry one, with phi_s = 1 - exp(-(Gamma + s) / Gamma_ref) the soap coverage from adsorbed and dissolved surfactant. On glass and glaze the three coefficients are 0.8 / 0.7 / 0.2 (repository pad value; an estimate in the 0.4-1.0 range found for wet fingers on glass; fingertip measurements on surfactant solution). Measured (P17): a glass pinched at 20 N per finger slips at the pinch under a 6.25 N pull when soapy, against 31 N when rinsed and 30.5 N when dry.

6. Numerics and cost (`world.py`, `water/system.py`)

6.1 Clocks and coupling

One tick is 0.5 ms. `pre_step` applies the water's forces of the last tick (impact, reaction, film shear, buoyancy and drag, the valve and sponge springs) to the bodies; MuJoCo steps; `post_step` emits the outlets' packets, advances the packets and pools every tick and the films on a stride chosen from the fastest cell's CFL number (substepping inside a film step when a jet's film is fast), reads back the pools' inflows and each body's film mass, and updates buoyancy every fourth tick and pad friction and the sponge every tenth. The coupling is explicit and one tick late, which the 0.5 ms step keeps well inside the stability of the heaviest water load on the lightest body: up to 12.8 kg pooled in the 4.5 kg can in the can rinse (A4), which runs with no fault.

6.2 Guards and invariants

The rigid world keeps `classroom_arrangement`'s guards (non-finite state, an object faster than its speed or spin limit, a solver warning) and reports them as faults, never hiding them. The water engine cannot create or lose water silently: every transfer subtracts and adds the same float64 number, and a packet older than 3 s, leaving the room or overwritten in the packet ring is counted as lost, by cause, in every verdict (across the eleven judged runs of the assisted and robot demonstrations none aged out and none was overwritten in the packet ring, and 11.6 mg left the room in the can rinse, A4). All three ledgers are a gate of every verdict.

6.3 Holding and resting

MuJoCo's soft friction lets a pinched glass creep down in the hand under its own weight (0.05 mm/s at 20 N per finger, P18), a glass slide down the drainboard's 0.3 deg pitch and a tomato roll on the colander's 0.6 deg floor, although each is below its friction or rolling threshold and a real one stays put. A post-solve no-slip pass removes the creep, at a cost of iterations x rows^2 per tick. Holding the glass (P18): with no pass it slides 0.16 mm in 3 s; `painting`'s setting (1000 iterations to 1e-15) costs 249 ms per tick among the pot sink's ~350 constraint rows (497x wall/sim; 49 ms and 98x at 300 iterations); `classroom_arrangement`'s (100 iterations to 1e-10) holds it to 0.0005 mm in 3 s at 11.5 ms per tick (23x), stops the resting glasses and slows the tomatoes to a steady 0.4 um/s (45 um in a 120 s rest, P11). The pass rewrites friction forces after the main solve and leaves the normal forces as solved. That is harmless at creep speeds and harmful in real sliding: with the pass on, a wet sponge pressed and slid over the pan's floor rocks and can sink through the pan's 4 mm floor hull. So the environment runs the pass at classroom_arrangement's setting while no contact slides faster than 5 mm/s, measured every 10 ticks, back on below 2 mm/s (`params.HOLD`, `world._noslip_schedule`). Creep is micrometres per second and scrubbing centimetres to decimetres. Gated this way, in the 120 s rests (P11) no glass moves more than 0.24 um and no tomato more than 45 um, and the scrub (A5 scrub) runs with no fault.

6.4 Checkpoints

`env.checkpoint()` copies the rigid state, controller state, MuJoCo's warm start and clock, the pad frictions, the no-slip pass's on/off state, the world's step-time flags, timers and filters (`env.STEP_STATE`: the hold flag, the slip monitor, the base's reflex hold, the IMU and glide filters), every GPU array of the water engine, the pools and outlets and the evaluator's counters; `restore` puts them back. Without the step-time state, an evaluator control (P10) that ends with a glass in the hand would leave the hold flag set through the restore that begins the next control, which would then fail `water_off_and_hands_free` as well as its own gate. A check with water running (P16) confirms the restore: the water account after a restore equals the snapshot's term by term, every plain attribute of the world (73 compared) equals its value at the snapshot, and a second branch from the same snapshot repeats the first. The evaluator's settle branch and `gar` reset depend on this.

7. Parameter ledger (`params.py`)

Every constant the water simulation reads is a ledger entry with value, unit, label and basis; `python -m surface_rinsing.params` prints the full table. Labels: *datasheet*, *regulation*, *standard*, *literature-read* (the source was read during the research), *search* (found, not read in full), *literature* (from knowledge of the field), *measured* / *derived* (in this repository), *repo* (inherited from a qualified environment), *estimate* (a reasoned number with no source for this object), *authored* (a design choice, mostly success thresholds), *calibrated* (fitted here on a stated observable). The ones that carry the tasks:

Quantity	Value	Basis
supply pressure, temperatures	60 psi, 60 C hot, 15 C cold (seeds: 40-80 psi, 5-25 C cold)	DOE / ASTM F2324 test pressure; ASHRAE via DOE; estimate
pre-rinse valve	1.15 gpm at 60 psi, nine jets, 28 m/s, 2.0 N	T&S B-0107 listings (search); orifice velocity from Bernoulli; the momentum flux reproduces DOE's force / flow line to 1.4 %
lever	13 N cracks, 22 N full, 10 mm; hinge 1e-3 kg m2 added inertia, critically damped (0.12 N m s/rad)	US 7,150,415 (3-5 lbf, literature-read); travel estimate; inertia numerical, damping derived from it (§5.1)
hose valve	3.89 gpm rated, 35 ft of 3/8 in hose -> 3.3 gpm, 24 m/s, 5.1 N at 60 psi (2.7 / 3.9 gpm at 40 / 80 psi)	T&S B-7132-01 (search); Blasius line loss (derived; M2)
swing faucet	2.2 gpm laminar regulator	federal maximum for kitchen faucets (regulation); regulator class estimate
splatter	xi(We, l/d), onset l/d = 130 / (1 + 5e-7 We^2), plateaus 0.30 / 0.37 / 0.75	Bhunia & Lienhard 1994 (literature-read, figures read from page images); droplet speed and angle estimates
lip separation	K = 0.44	Duez et al. 2010 as closed in `wine_pouring` (repo)
wash solution	0.75 g/L surfactant (seeds 0.5-1.25), CMC 0.42 g/L	dosing from product directions (search); LAS CMC (literature)
adsorption	Gamma_m 0.1-0.5 mg/m2 by substrate, k_d 0.05 /s	estimate bracketing measured residues of 2-7 ug per dish (search)
foam	5 % liquid, yield 5 Pa, half-life 300 s	Mason et al. (literature); half-life estimate
contact angles, bead retention	glass 20, glaze 25, steel 70, LLDPE 95, tomato skin 100 deg; 0.02-0.10 mm on a vertical wall	literature orders; canopy interception storage ~0.05 mm per leaf area (search) and the pinned-drop balance (estimate)
tomato paste	0.20 kg/m2 dried, sigma_dry 200 kPa, sigma_wet 150 Pa, t_soak 5.9 s at 20 C	F2324-03 recipe (standard, via programme documents); yield stress order of tomato paste (literature); t_soak calibrated on bench B6
dried egg, dust	300 Pa / 40 s; 0.5 Pa / 2 s	estimates
cleaning-front constant	k' 2e-3 s m/kg	Wilson et al. 2014 range 1e-4 to 2e-3 (literature-read)
sponge	111 x 66 x 17 mm, E 50 kPa over 30 cm2, 79 mL of pores, 6 mm soft margin, abrasion constants	product listing (search); wet cellulose 20-200 kPa (search); the margin from that modulus (§5.3, estimate); abrasion estimates
pad friction on glass	0.8 dry, 0.7 wet, 0.2 soapy	repo; wet-finger range (search); fingertip on surfactant solution (search)
success thresholds	10 ug per dish, 1 mg/m2 gauge, 1 ug/cm2 visible soil, 30 s sanitizer	measured residues after ordinary rinsing 2.1-7.2 ug (search); visible-residue limit (search); 30 s is a posted contact time, not the code minimum for this bath: Food Code 4-703.11(C) asks 10 s for a 100 ppm chlorine solution under 4-501.114(A) and 30 s for other chemical sanitizers (regulation); gauge size authored

What is an estimate stays an estimate: the splatter droplets' speed and angle, every soil constant except the paste's soak time, the sponge's abrasion constants, bead retention depths, the pool exchange rate and the foam half-life have no measurement behind them for these objects.

8. Demonstrations

Each assisted demonstration (A1-A6) carries out the interactions of one task with labelled phantoms: a six-axis servo on a body with its weight compensated and its force capped at 60 N (the phantom hand), a PD torque capped at 1 N m on a wrist blade (phantom fingers), and equal and opposite forces on a spray valve's lever pad and body (the phantom squeeze). The phantoms never move water, soap or soil. The valves open by their joint angles, and everything the water does follows from that. In the robot demonstrations (D1, D2) a scripted operator drives the robot through its interface. Every run ends with the evaluator's verdict on the whole task. Each contact sheet shows twelve frames of the run's captioned video, rendered from the recording of the run that produced the numbers. The maps are drawn by the demonstration from the evaluator's own fields.

Run	Task	Assistance	Simulated (s)	Evaluator's verdict	Protective stops
A1	rinse-glassware	phantom hand on the glass, phantom fingers on the faucet blades	54.6	fails soap_free, sanitized, placed	0
A2	pre-rinse-plates	phantom hand and squeeze on the spray valve, phantom fingers on the supply blades	41.5	fails visibly_clean	0
A3	rinse-tomatoes	phantom hand on the colander, phantom fingers on the faucet blades	60.0	fails visibly_clean (two repeat runs pass every gate)	0
A4	rinse-garbage-can	phantom hand and squeeze on the hose valve, second phantom hand on the can	128.6	every gate passes	0
A5 rinse	rinse-sponge	phantom hand and squeeze pair on the sponge, phantom fingers on the faucet blades	20.3	every gate passes	0
A5 scrub	scrub-rinse-pan	phantom hand on the sponge, second phantom hand steadying the pan by its handle	80.1	fails visibly_clean, soap_free, sanitized, drained, placed	0
A6	flush-floor	phantom hand and squeeze on the hose valve	221.8	every gate passes	0
D1	pre-rinse-plates	the robot, with phantom fingers opening the supply blades in the first second	28.0	fails visibly_clean	0
D2	rinse-sponge	the robot, no assistance	64.5	fails no_suds, wrung_out	6

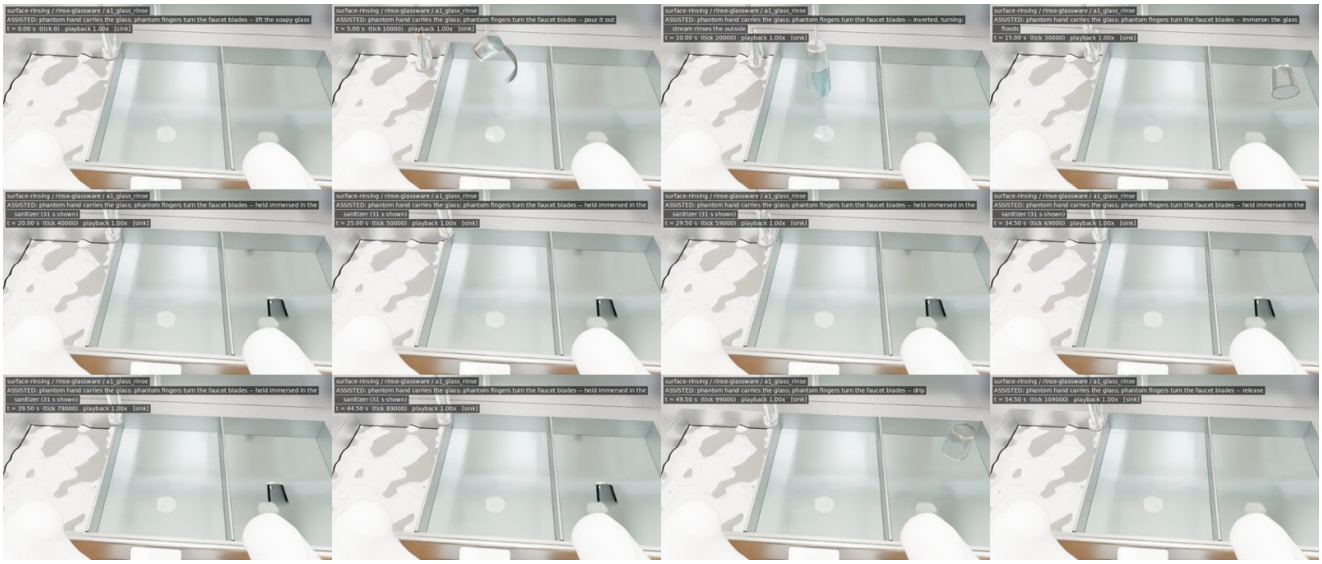


Figure 7. A1, rinse-glassware, sink camera. ASSISTED: phantom hand on the glass, phantom fingers on the faucet blades. The soapy glass (1.61 mg of detergent) is filled through its mouth under the faucet until it overflows and is poured out, rinsed again tilted and poured out, then inverted and turned under the stream. It is dunked in the sanitizer, where it floods, held there 31.55 s, lifted out mouth-down to drain and set mouth-down in the glass rack. At the verdict it carries 0.637 ug of detergent (limit 10 ug) and every gate on it passes. The task fails soap_free, sanitized and placed on the five glasses the run leaves on the drainboard.

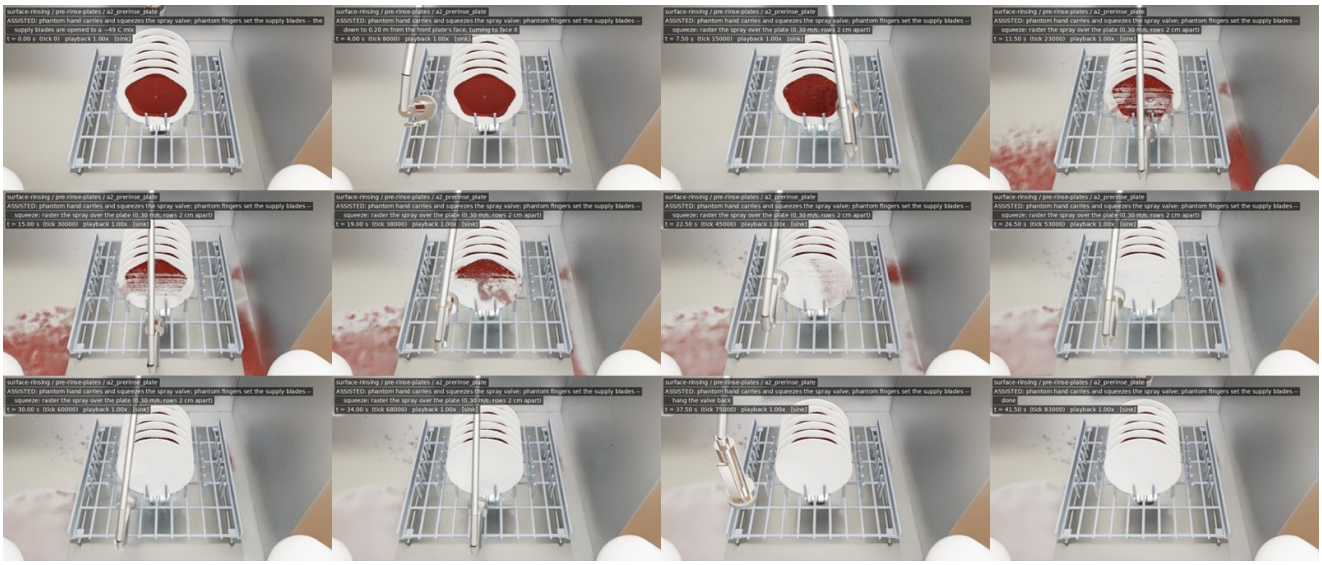


Figure 8. A2, pre-rinse-plates, sink camera. ASSISTED: phantom hand and squeeze on the spray valve, phantom fingers on the supply blades. The valve is brought to 0.20 m from the front plate, squeezed (1.147 gpm, jets at 28.1 m/s, reaction up to 2.02 N, water mixed to 48.9 C) and rastered over the face from 5.0 to 34.9 s. The paste soaks first and then comes off under the jets: 5073 mg on the front plate at the start, 1072 mg at 20 s, 34 mg at 22.5 s and none from 30 s. No gauge on it is visibly soiled after 25.7 s of spray (FSTC verified valves: 16-26 s). The other five plates are not sprayed, so the task fails visibly_clean.

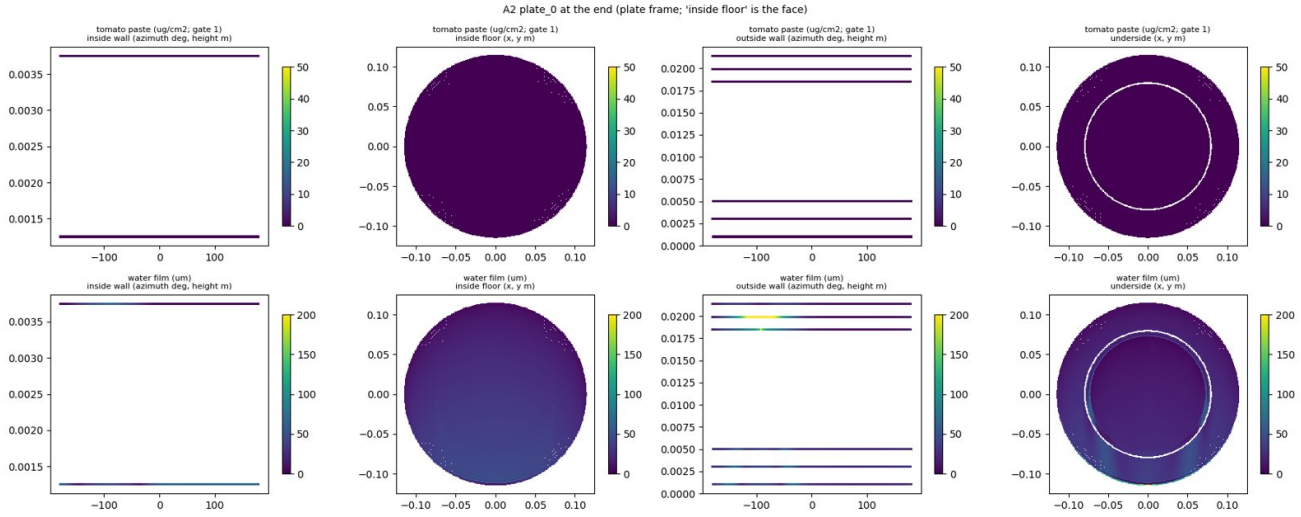


Figure 9. A2: the front plate at the end, unrolled in the plate's frame. Tomato paste (top row, ug/cm2, gate 1) and water film (bottom row, um) on the inside wall, the inside floor (the face), the outside wall and the underside. No paste is left on any face.

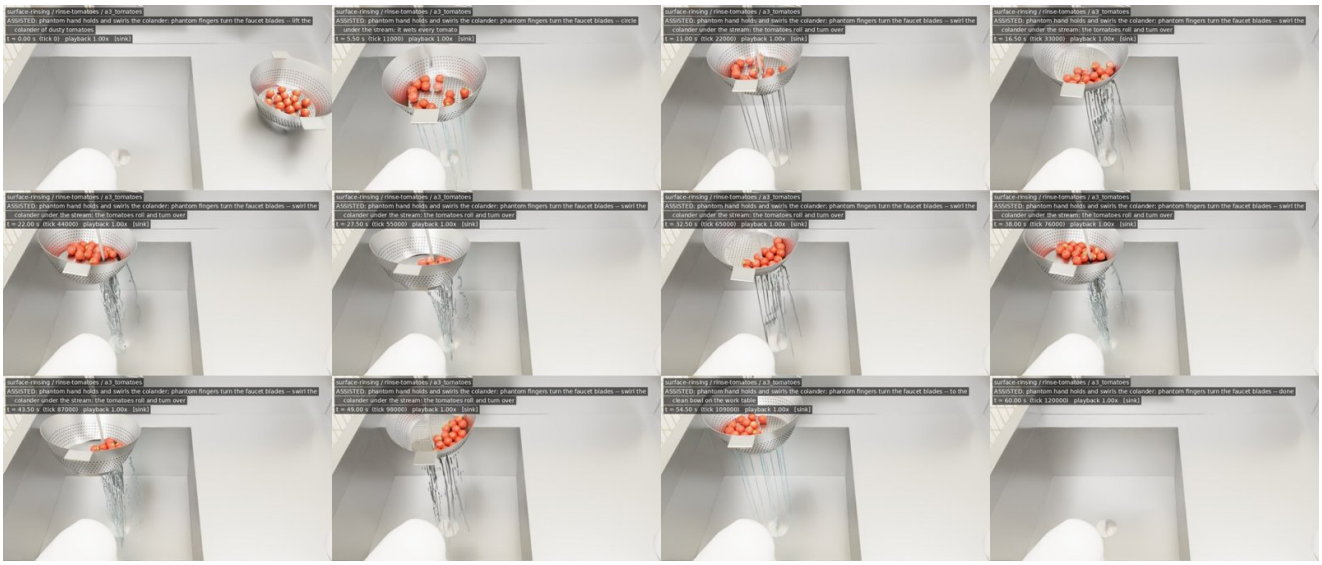


Figure 10. A3, rinse-tomatoes, sink camera. ASSISTED: phantom hand on the colander, phantom fingers on the faucet blades. The colander is carried under the stream, circled so the stream wets every tomato, swirled so the tomatoes roll and turn over, drained over the sink and set in the clean bowl. The tomatoes are free bodies and float in the colander's pool (up to 27.6 mL) when the holes cannot keep up. Dust on them falls from 14.4 mg to 0.025 mg. One tomato keeps a 1 cm gauge at 21.3 ug/cm2 (gate 1), so this run fails visibly_clean. Two more runs of the same plan end with no dust and pass every gate. No tomato left the colander in any of the three.

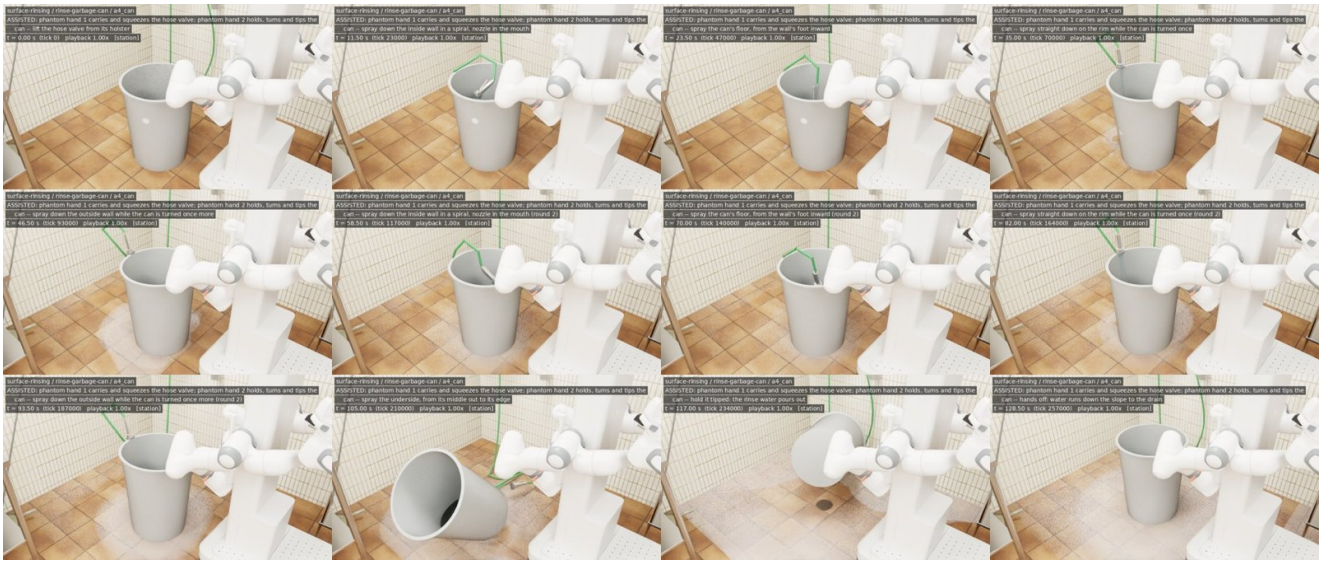


Figure 11. A4, rinse-garbage-can, bay camera. ASSISTED: phantom hand 1 carries and squeezes the hose valve, phantom hand 2 holds, turns and tips the can. Twice over, the nozzle in the can's mouth sprays the inside wall in a spiral and then the floor, and the jet goes down on the rim and the outside wall while the can is turned. The can is leaned back 45 deg for the underside, the valve is docked, and the can is tipped to 105-107 deg toward the floor drain so the rinse water pours out, then stood up. Flow 3.33 gpm at 24.2 m/s, reaction up to 5.04 N, up to 12.8 L pooled in the can. Detergent on and in the can falls from 56.5 mg to 0.148 mg and debris from 354 mg to none. Every gate passes.

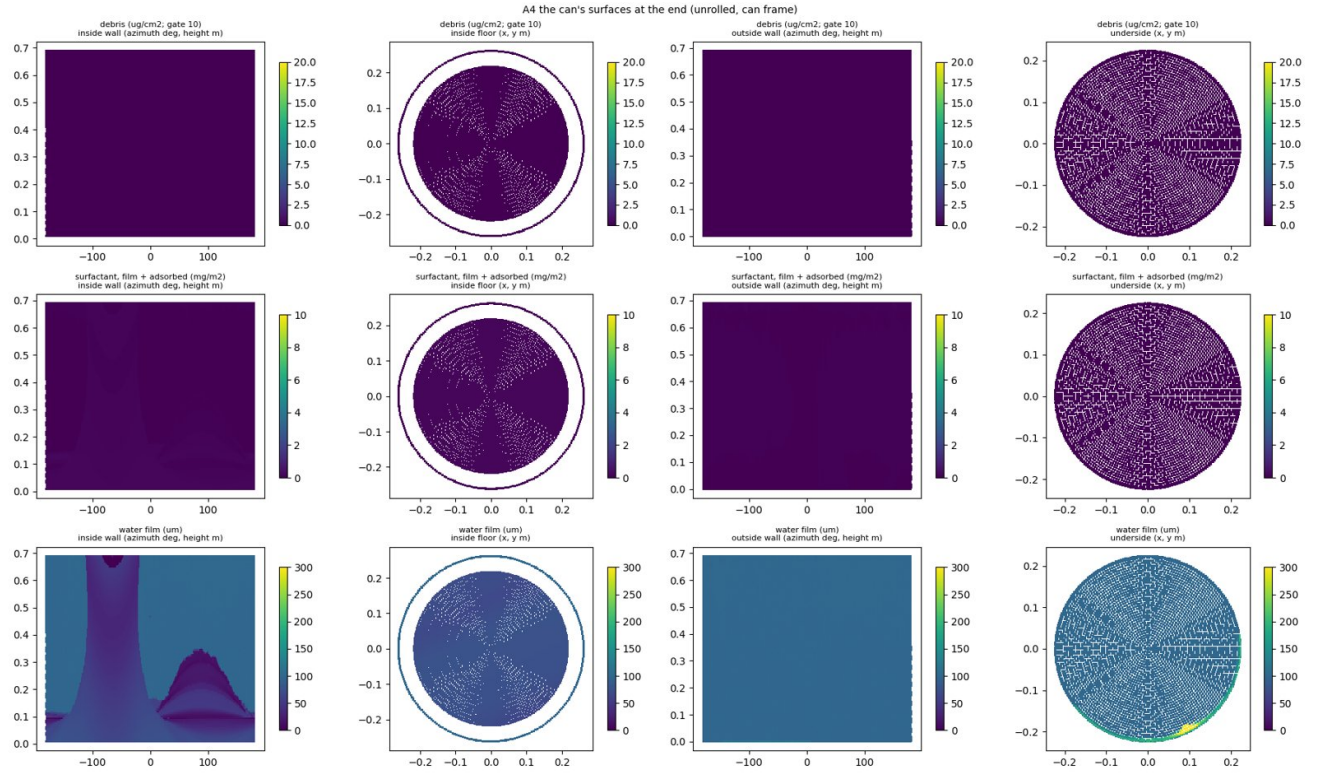


Figure 12. A4: the can's surfaces at the end, unrolled in the can's frame. Debris (ug/cm2, gate 10), surfactant in the film and adsorbed (mg/m2) and water film (um), on the inside wall, the inside floor, the outside wall and the underside.

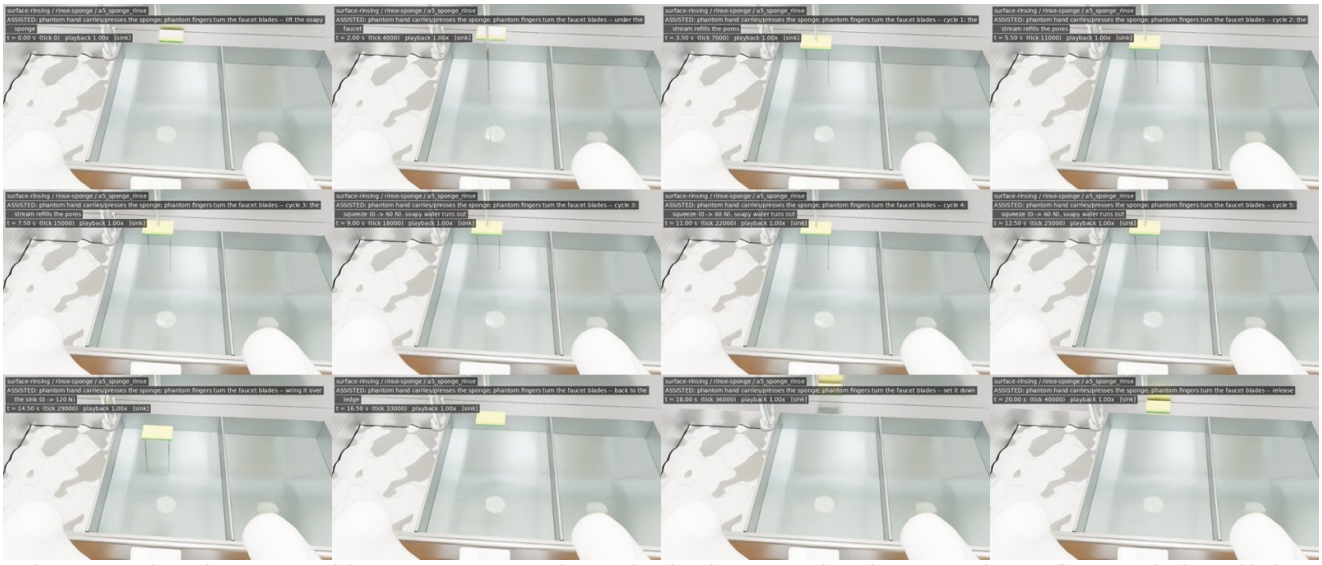


Figure 13. A5 rinse, rinse-sponge, sink camera. ASSISTED: phantom hand and squeeze pair on the sponge, phantom fingers on the faucet blades. The soapy sponge (59.3 mL of 0.75 g/L wash solution in its pores) is held under the faucet. Five times the stream refills the pores and a 60 N squeeze (31 % strain) expels the water above the squeezed capacity. The pore concentration falls to 0.0058 g/L after the first squeeze and 0.00007 g/L after the second. With the faucet off, a 120 N wring over the sink leaves 41.4 mL (52 % of capacity), and the sponge is set back on its ledge. Every gate passes.



Figure 14. A5 scrub, scrub-rinse-pan. DIAGNOSTIC render: the wash bowl's water and suds are not drawn, since they hide the pan from every camera, and they are still simulated. ASSISTED: phantom hand on the sponge, second phantom hand steadying the pan by its handle. As a control the strokes first run 3 mm above the soil, and no egg comes off. Three passes over the middle at 15 N, the sponge turned between them, take the egg from 2797 mg to 1755 mg. An orbit round the floor's edge and the foot of the wall at 25 N takes it to 230 mg. The run covers the scrub only, so the task fails visibly_clean (worst gauge 5621 ug/cm2), soap_free, sanitized, drained and placed.

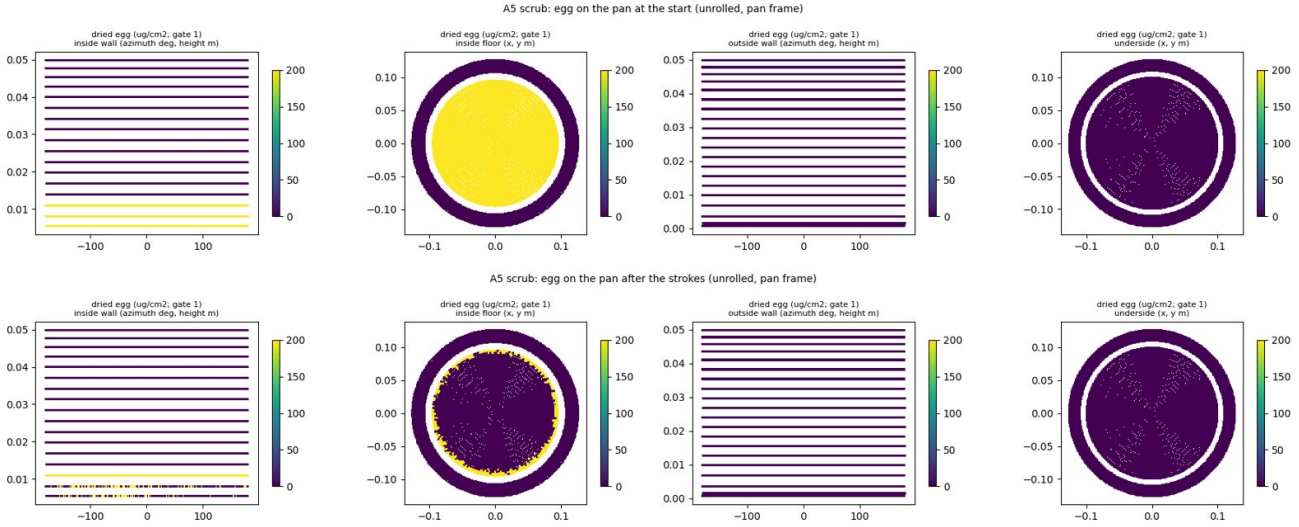


Figure 15. A5 scrub: dried egg on the pan (ug/cm2, gate 1), unrolled, at the start (top) and after the strokes (bottom). The middle of the floor is clean. A band at the floor's edge and the foot of the wall keeps egg.

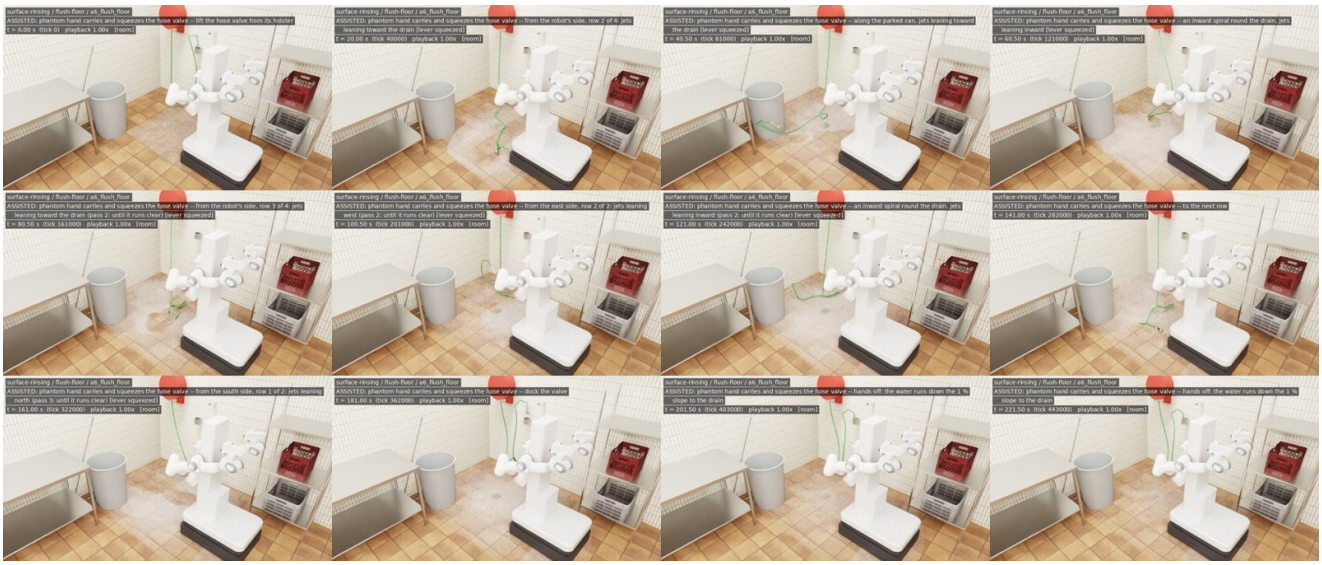


Figure 16. A6, flush-floor, room camera. ASSISTED: phantom hand and squeeze on the hose valve. The hose valve is lifted from its holster, and the floor is worked from the robot's side, in rows toward the drain and in an inward spiral round it, over repeated passes with the jets leaning toward the drain. The valve is docked and the floor drains hands-off. Flow 3.33 gpm at 24.2 m/s, reaction up to 5.04 N. Debris on the floor falls from 1128 mg to 2.7 mg and 22.25 L go down the drain. At the verdict the worst 1 cm gauge is 2.762 ug/cm2 (gate 10) and 0.493 mg of detergent is left. Every gate passes. The evaluator's settle stopped at its 30 s limit with water still running off the floor.

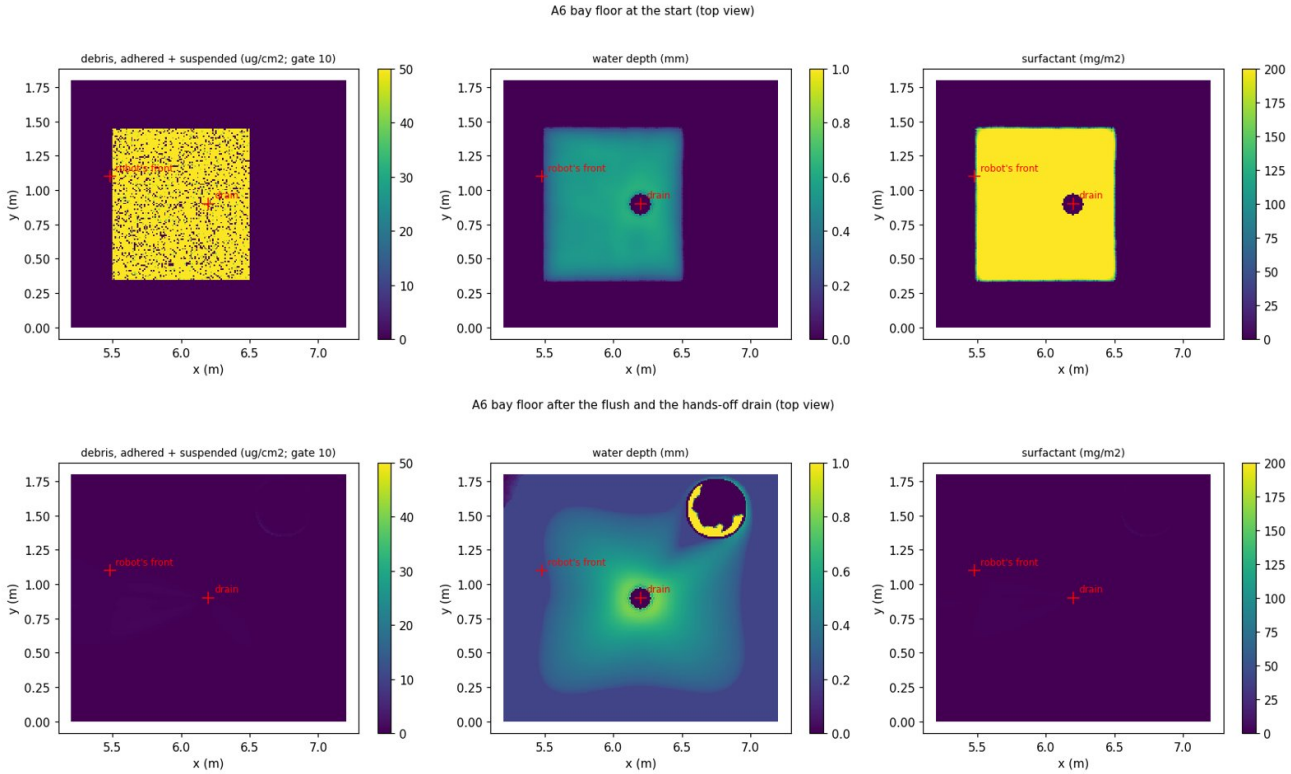


Figure 17. A6: the bay floor at the start (top) and after the flush and the hands-off drain (bottom), top view. Debris (ug/cm2, gate 10), water depth (mm) and surfactant (mg/m2), with the drain and the robot's front marked.

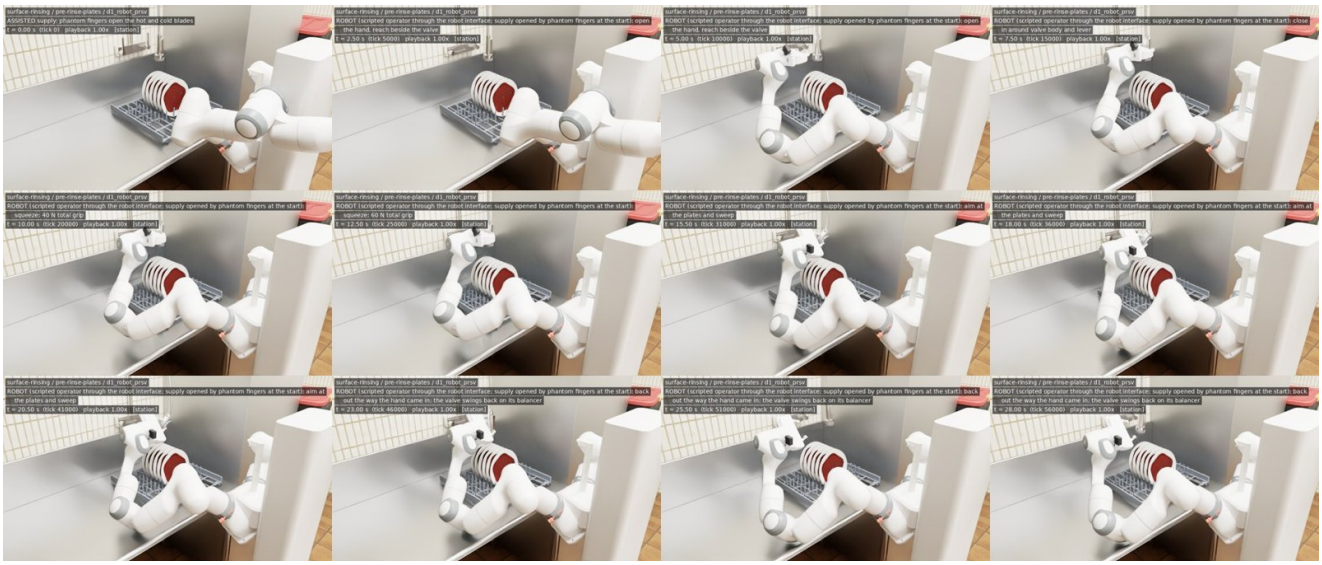


Figure 18. D1, pre-rinse-plates, station camera. The robot squeezes the hanging spray valve, driven by a scripted operator through the robot interface. Phantom fingers open the supply blades in the first second, and nothing else is assisted. The hand reaches beside the valve, closes round the valve body and the lever, and squeezes in steps. 20 N of total grip does not crack the lever, 40 N opens the valve to 71 % (0.83 gpm, 20.3 m/s) and 60 N opens it fully (1.147 gpm, 28.1 m/s, reaction 2.02 N). The hand aims the spray at the plates and sweeps for 7 s, lets go so the valve snaps shut, and backs out. The valve swings back on its balancer and ends 4.3 cm from its parked pose. No protective stop. Every plate is still visibly soiled, so the task fails visibly_clean.

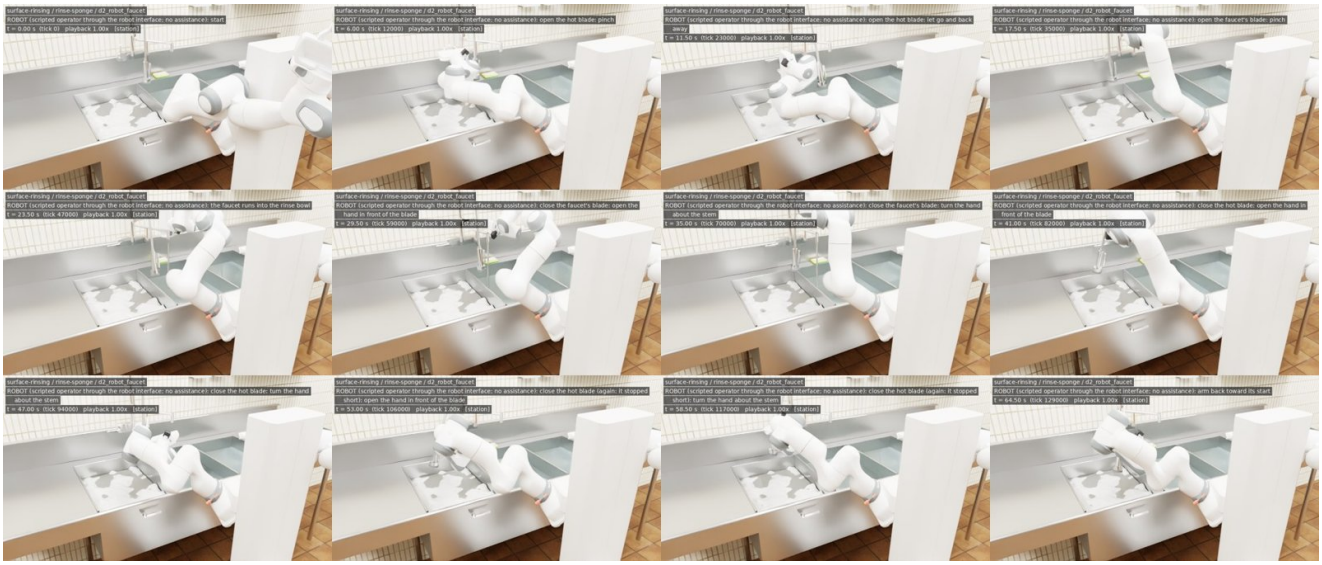


Figure 19. D2, rinse-sponge, station camera. The robot turns the water on and off with its fingers, driven by a scripted operator with no assistance. The water engine runs on the CPU. The left hand pinches the hot blade and turns it 90 deg, then turns the swing faucet's blade to 88 deg: the faucet runs into the rinse bowl at 2.20 gpm and 60 C. It turns the faucet's blade back to 0 and closes the hot blade, which takes a second try after the first stops short. The run latched 6 protective stops. It ends with every blade at 0 deg, the water off and the hands free, so water_off_and_hands_free passes. The sponge is not touched, so the task's no_suds and wrung_out gates fail.